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CS101C: Introduction to Programming (Using C)

Instructors: Prof. Aikta Arya and Prof. Konjengbam Anand

CSE Department

Indian Institute of Technology, Dharwad

Week 8

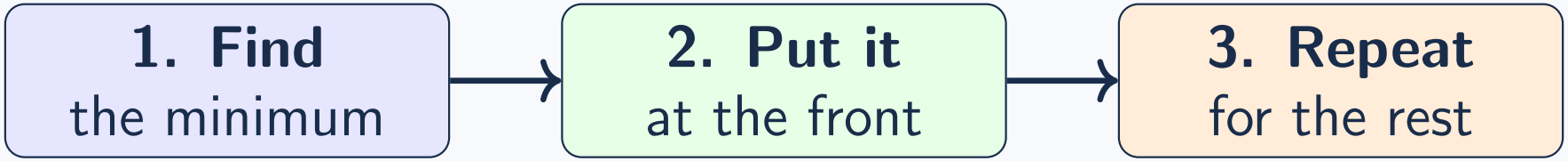
Today's Class (28/9/2026)

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"Find the smallest and put it in front."

How does Selection Sort work?

Repeatedly:



Find minimum $\longrightarrow$ Place it $\longrightarrow$ Repeat

SELECTION SORT

॥ सा विद्या या विमुक्तये ॥



SELECTION SORT

॥ सा विद्या या विमुक्तये ॥



SELECTION SORT

॥ सा विद्या या विमुक्तये ॥



SELECTION SORT



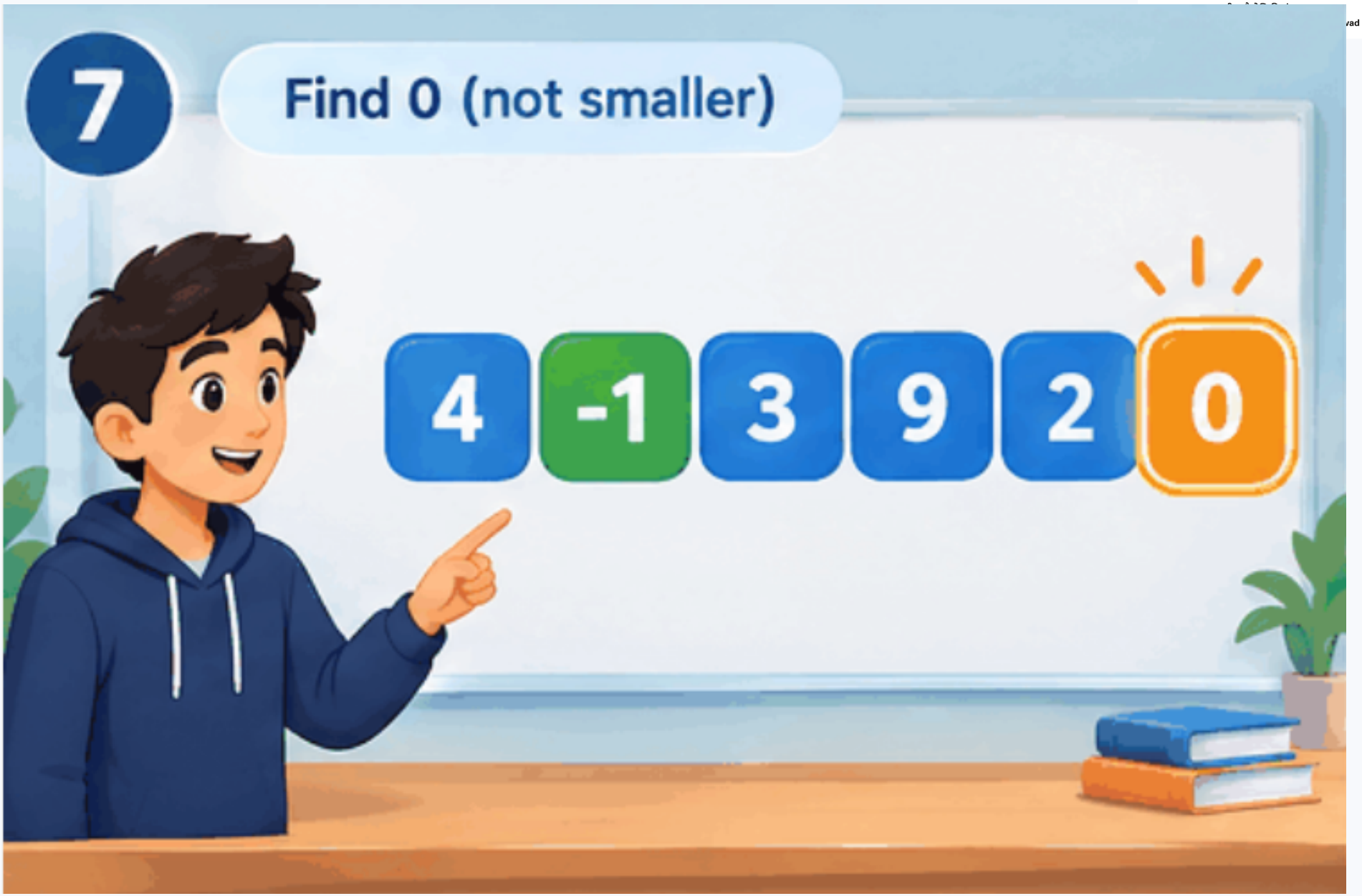
SELECTION SORT



SELECTION SORT

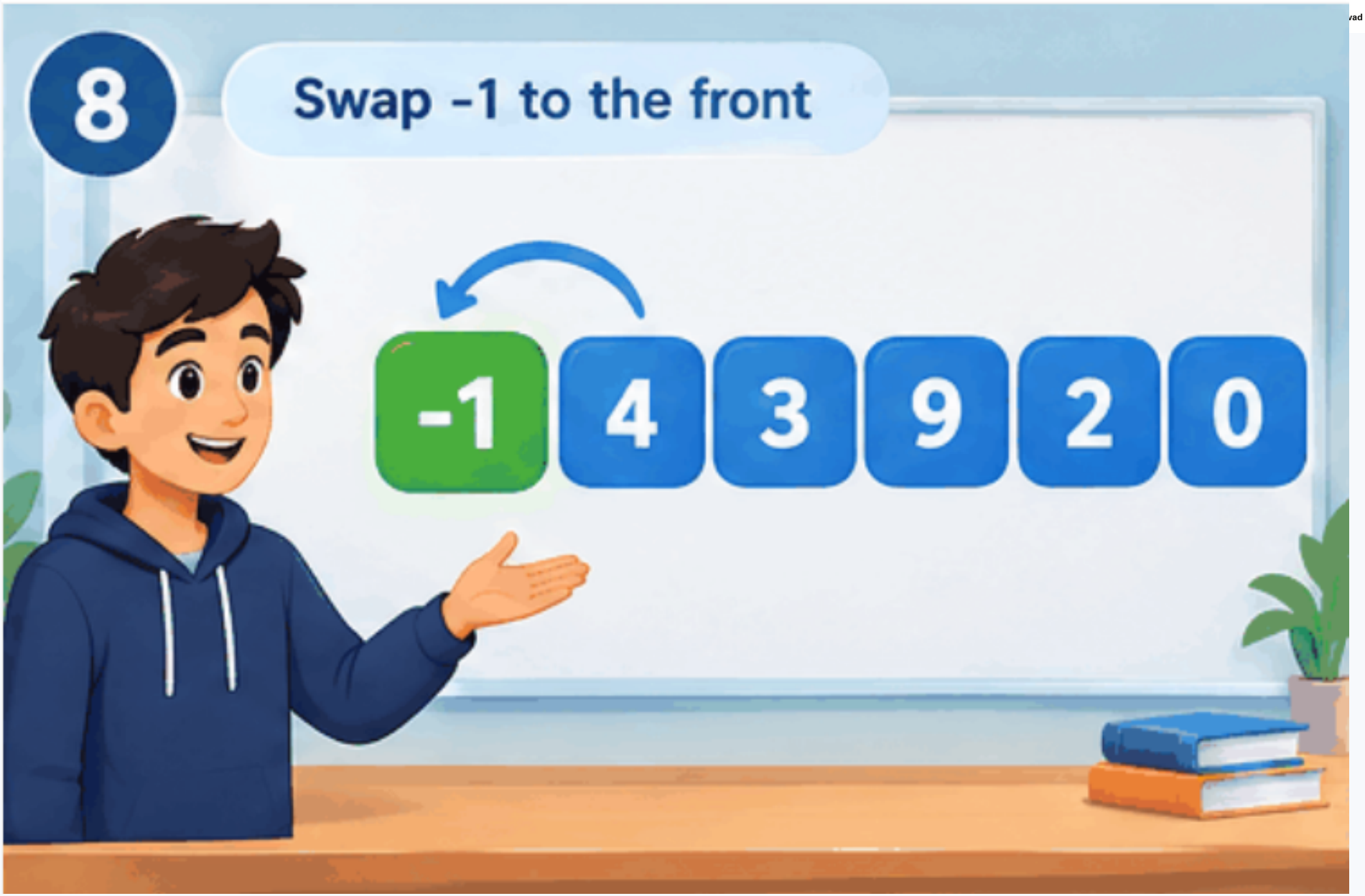


SELECTION SORT



SELECTION SORT

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SELECTION SORT



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We already know WHAT to do...

Now let's tell the computer HOW to do it.

Choose a position → **Find minimum** → **Swap**

Outer loop

Choose the position

where the next smallest element should go.

Inner loop

Search REST

to find the smallest element.

Two different jobs $\Longrightarrow$ Two loops

Selection Sort — Pseudocode

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Now let's put the whole idea together.

```
SelectionSort(input, N)

for cursor = 0 to N-1

    smallest = cursor

    for i = cursor to N-1

        if input[i] < input[smallest]

            smallest = i

    swap(input[cursor], input[smallest])
```

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Think!

Outer loop

Where should the next smallest element go?

Inner loop

Which element is the smallest in REST?

Swap

Put the smallest element at the cursor.

OUTER LOOP $\rightarrow$ choose position

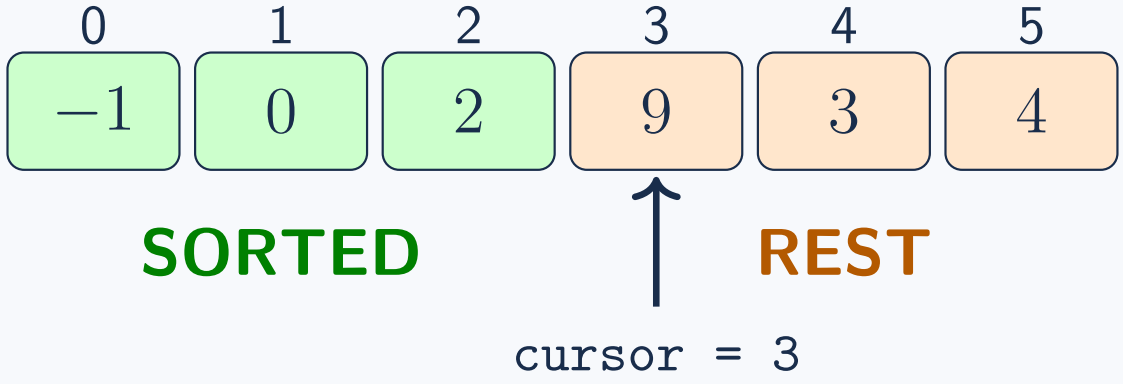
INNER LOOP $\rightarrow$ find minimum

SWAP $\rightarrow$ place minimum

What Does cursor Mean?

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The cursor is the boundary between SORTED and REST.



When `cursor = 3` we have:

Already sorted

$[0,$ `cursor`$)$

indices $0, 1, 2$

Still to process

$[$`cursor`$, N)$

indices $3, 4, 5$

`[0, cursor)` means:

start at 0, stop BEFORE cursor

Why Is It Called "Selection" Sort?

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What are we selecting?

The smallest element



Put it at the front

SELECT $\longrightarrow$ **PLACE** $\longrightarrow$ **REPEAT**

Remember

We **select the minimum** from REST
and place it at the beginning of REST.

How Much Work Does Selection Sort Do?

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$$\boxed{N = 6}$$

Round 1: 6 elements

Round 2: 5 elements

Round 3: 4 elements

Round 4: 3 elements

Round 5: 2 elements

Round 6: 1 element

Total search:

$$6 + 5 + 4 + 3 + 2 + 1$$

$$= \frac{6(7)}{2} = 21$$

For general N:

$$\boxed{N + (N-1) + (N-2) + \cdots + 1 = \frac{N(N+1)}{2}}$$

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What is the inner loop actually doing?

It searches REST to find the minimum.

Search REST → **Find minimum**

The amount of work grows approximately as:

$$\frac{N(N+1)}{2} \approx \frac{N^2}{2}$$

Ignoring constants:

$$O(N^2)$$

Interactive Question — What If We Double N?

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Suppose

$$N = 100$$

Roughly:

$$N^2 = 100^2$$

$$\boxed{10,000}$$

Now double it!

$$N = 200$$

Roughly:

$$N^2 = 200^2$$

$$\boxed{40,000}$$

Did the input double? Yes.
Did the work double? No!

$$\boxed{40,000 = 4 \times 10,000}$$

Key idea

Doubling the input makes quadratic work roughly 4×.

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Is "work" only about comparisons?

Comparisons

Search the REST part.

$$\boxed{O(N^2)}$$

Many comparisons!

Data Movement

Place the selected minimum.

$$\boxed{O(N) \text{ swaps}}$$

At most one swap
per outer-loop iteration.

Comparisons: $O(N^2)$

Swaps: $O(N)$

Big Takeaway

Selection Sort does many comparisons, but relatively few swaps.

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What should we remember?

SELECT minimum → **PLACE at cursor** → **REPEAT**

Property	Selection Sort
Comparisons	$O(N^2)$
Swaps / data movement	$O(N)$
Extra memory	$O(1)$

One sentence

Find the minimum, put it at the cursor, and repeat.

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What if the array is already sorted?



Can we stop now?

The surprising answer

No! Selection Sort still searches REST.

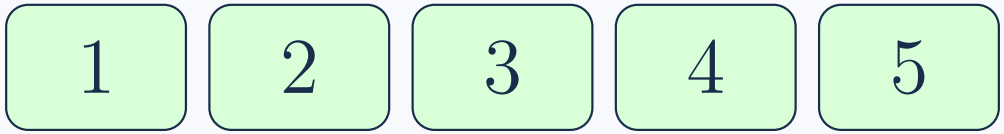
Case	Time
Best case	$O(N^2)$
Average case	$O(N^2)$
Worst case	$O(N^2)$

Why?

Selection Sort does **not** have an early-stop condition.
It keeps searching the remaining array on every pass.

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What if the array is already sorted?



Will Selection Sort finish immediately?

A. Yes — it detects that the array is sorted.
B. No — it still searches REST on every pass.
C. It depends on the input.

Think before answering!

Does our Selection Sort algorithm have an early-stop condition?

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B. No!



Still searches the remaining array!

$$\boxed{\text{Best Case} = O(N^2)}$$

$$\boxed{\text{Average Case} = O(N^2)}$$

$$\boxed{\text{Worst Case} = O(N^2)}$$

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How many comparisons?

Suppose:

$$\boxed{N = 100}$$

How many **actual element-to-element comparisons** does Selection Sort perform?

A. 100
B. $100^2 = 10,000$
C. $\dfrac{100(99)}{2}$
D. $\dfrac{100(101)}{2}$

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Selection Sort must find the smallest element

candidate

4	−1	3	9	2	0

The computer asks:

$-1 < 4\,?$ $\quad$ $3 < -1\,?$ $\quad$ $9 < -1\,?$

Element-to-element comparison

We compare the value of **one array element** with the value of **another array element**.

This is an actual comparison.

Challenge 2 — Answer

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Answer: C

The comparisons are:

$$99 + 98 + 97 + \cdots + 2 + 1$$
$$= \frac{100(99)}{2} = \boxed{4950}$$

$$\boxed{\text{Actual comparisons} = \frac{N(N-1)}{2}}$$

The inner loop may execute

$$\frac{N(N+1)}{2}$$

times if it starts at $i = \texttt{cursor}$, but the first element is not compared with itself.

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Can we remove the `swap`?

A student writes:

```
smallest = cursor;
for (i = cursor + 1; i < N; i++)
{
    if (input[i] < input[smallest])
        smallest = i;
}
input[cursor] = input[smallest];
```

Is this correct?

A. Yes
B. No
C. Only for sorted arrays

Think!

What happens to the **old value** at `input[cursor]`?

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B. No!

Suppose:

$$|4|\ -1\ |3|9|2|0|$$

The minimum is -1.
If we simply write:

```
input[0] = input[1]
```

we get:

$$|-1|\ -1\ |3|9|2|0|$$

What went wrong?

The original 4 has been **lost**!

Today's Class (30/9/2026)

You Have Already Used Functions!

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You have been using functions from Day 1!
You just didn't call them "functions" yet.

You have already written

```
main()
printf()
scanf()
pow()
```

For example

```
printf("Hello");

scanf("%d", &x);

y = pow(2,3);
```

WAIT!

Who wrote the code inside

`printf()`

You?

NO!

Someone had already written it.

You simply

CALLED the function

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Think of a function as a MACHINE!

You give it something → It does some work → You may get something back.



Let's see a real C function:

```
pow(2,3)
```

INPUT	FUNCTION	OUTPUT
$2,\ 3$	`pow()`	**8**
(base and exponent)	does the calculation	

INPUT → FUNCTION → OUTPUT

Why Do We Need Functions?

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Imagine calculating squares again and again...
Suppose we want the square of several numbers.

Approach 1 — Without a Function

```
int a = 5, b = 8, c = 12;
printf("%d\n", a*a);
printf("%d\n", b*b);
printf("%d\n", c*c);
```

Same logic $\rightarrow$ repeated again and again!

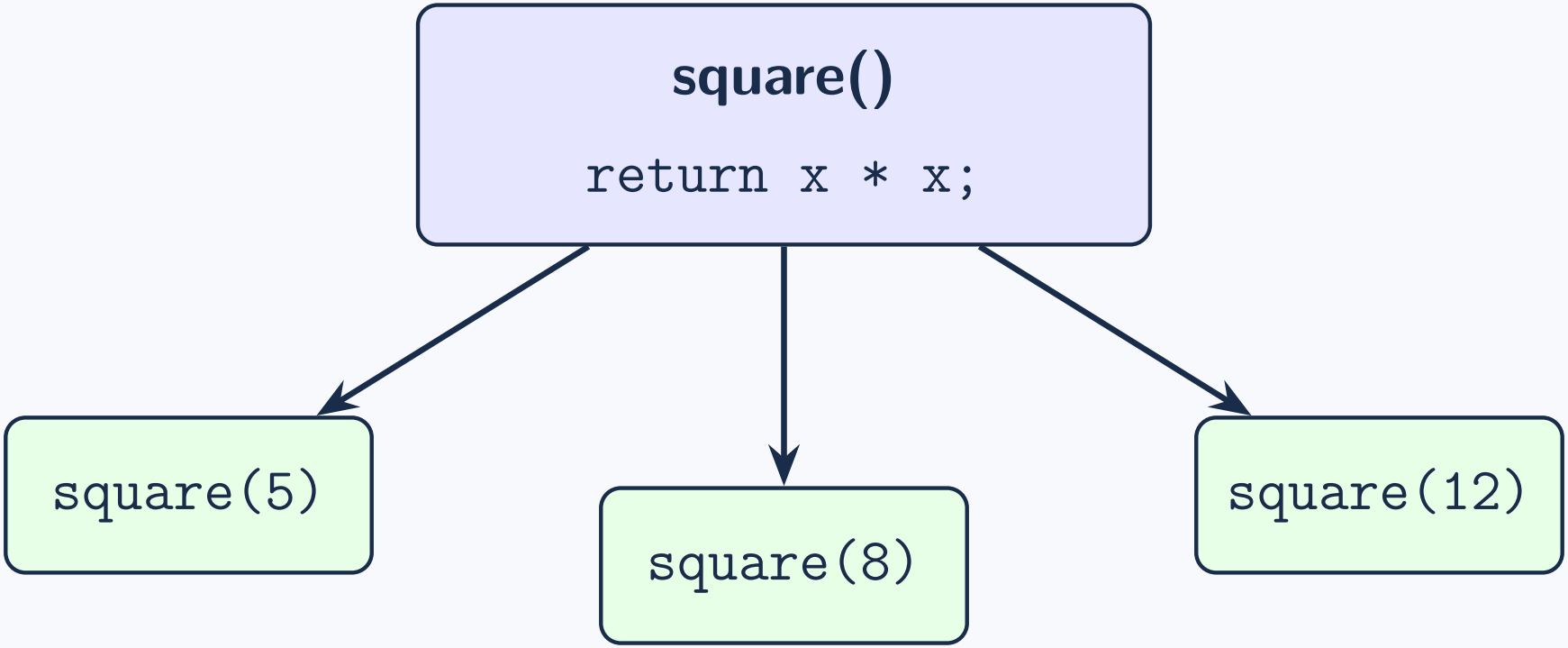
Can we write the logic only ONCE?



Why Do We Need Functions?

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Write the logic ONCE.
Then use it whenever you need it.




```
int square(int x)
{
    return x * x;
}
```

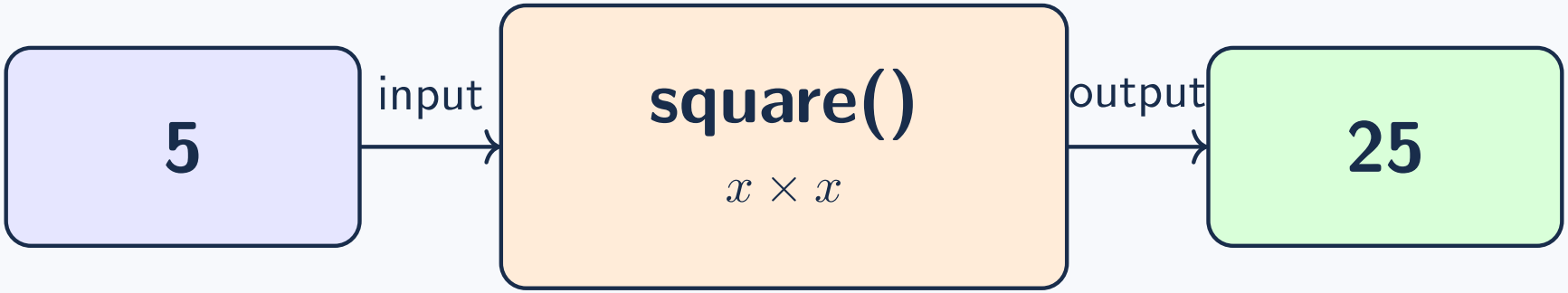
Write once → Use many times

Our First Function

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Let's build our first function!

We want a machine that calculates a square.




```
int square(int x)
{
    return x * x;
}
```

Now call the function:

`square(5)`

The function performs:

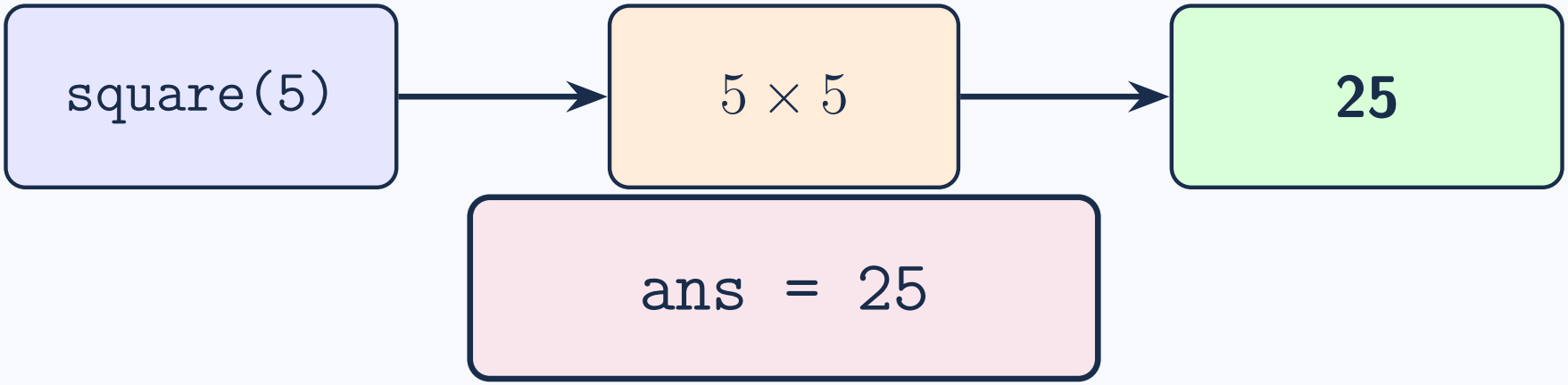
$$5 \times 5 = \boxed{25}$$

What If We Want the Answer?

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A function can give a value back to us.
We can store that returned value in a variable.

```
int ans = square(5);
```



Function gives 25 → variable stores 25

Understanding the Function

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Look carefully at this line!

```
int    square    (int x)
```

`int`	`square`	`(int x)`
Return Type	**Function Name**	**Parameter**
What type of value will come back?	The name we use to call the function.	The input received by the function.
`int` ↓ tells the type of value returned	`square` ↓ name used to call the function	`(int x)` ↓ input received by the function

Put the Three Parts Together

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Every function declaration tells us three things.

```
int    square    (int x)
```

int	square	(int x)
RETURN TYPE	**NAME**	**PARAMETER**

```
int square(int x)
{
    return x * x;
}
```

Think of it as:

WHAT COMES BACK? WHAT IS ITS NAME?
WHAT INPUT DOES IT TAKE?

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Function Definition

```
int square(int x)
{
    return x * x;
}
```

x

x = PARAMETER

A parameter is a **placeholder** inside the function definition.

Now suppose we call:

```
square(10);
```

⇓

10

10 = ARGUMENT

Parameter vs Argument

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PARAMETER
Placeholder
inside the function

→

ARGUMENT
Actual value
given during the call

```
int square(int x)          // x = parameter
square(10);                // 10 = argument
```

PARAMETER = Placeholder

ARGUMENT = Actual value

Another Example

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Let's try two parameters!

```
int add(int a, int b)
{
    return a + b;
}
```

a — Parameter

b — Parameter

Now call:

```
add(10,20);
```

10 — Argument

20 — Argument

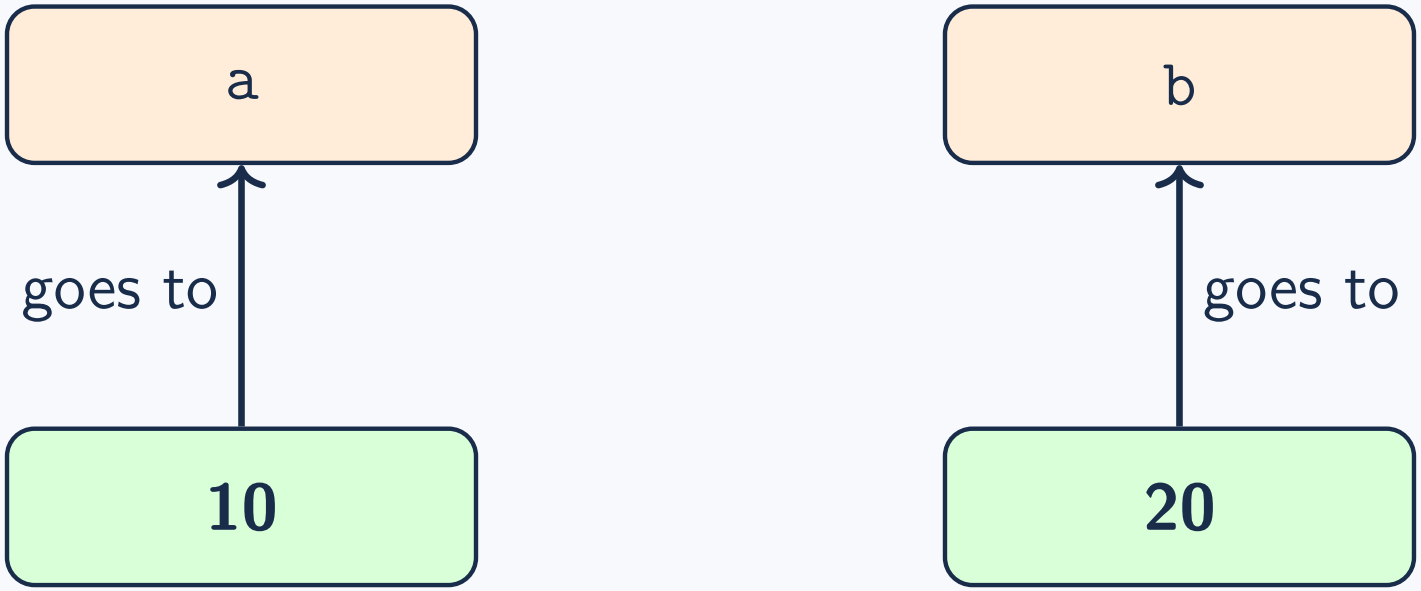
The function calculates:

$$10 + 20 = \boxed{30}$$

What Happens to the Values?

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Arguments are matched with parameters



Therefore:

`return a + b` $\Longrightarrow$ 30

What Is a Function Call?

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What actually happens when we call a function?

```
int answer;
answer = square(5);
```

```
answer = square(5);
```

Step 1: The computer reaches

```
square(5)
```

⇓

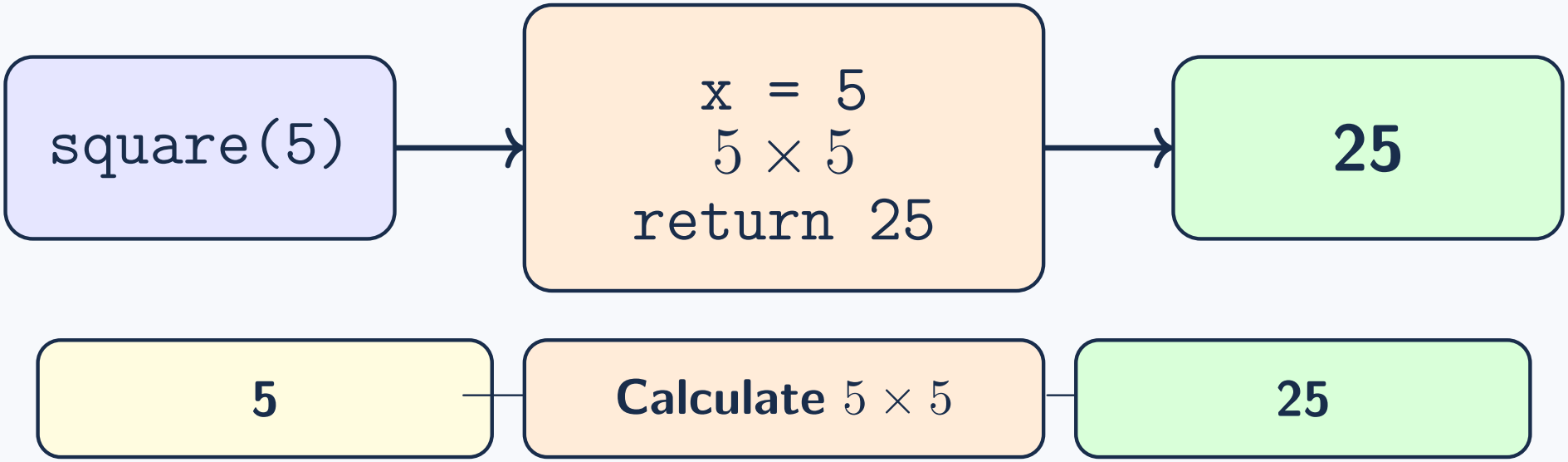
Step 2: It enters the function

```
int square(int x)
{
    return x * x;
}
```

The Journey of a Function Call

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Follow the value: 5




```
answer = square(5);
answer = 25;
```

Function Call = Call Site

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Where is the function being used?

```
int answer;
answer = square(5);
```

`square(5)`

This is where we **call** the function.

Function Call

The place where we use a function is called the **function call** or **call site**.

Remember the journey:

`square(5)` $\longrightarrow$ `x = 5` $\longrightarrow$ 25 $\longrightarrow$ `answer = 25`

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Let's Trace a Function!

```
int square(int x)
{
    return x*x;
}

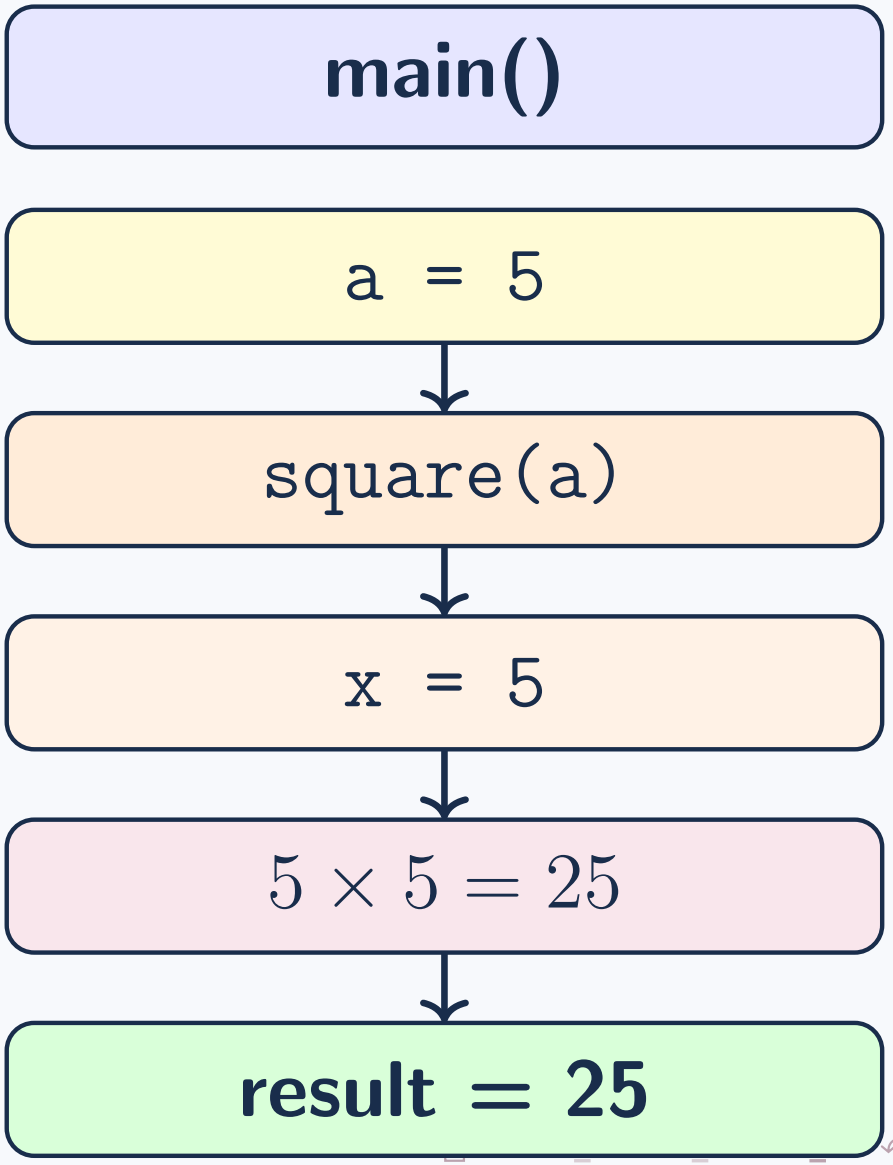
int main()
{
    int a = 5;

    int result = square(a);

    printf("%d", result);

    return 0;
}
```

Step-by-step execution



Question

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What will be printed?

```
int square(int x)
{
    return x*x;
}
int main()
{
    printf("%d ", square(3));
    printf("%d ", square(5));
    printf("%d", square(10));
    return 0;
}
```

Don't calculate everything at once!

`square(3)` → `square(5)` → `square(10)`

A. 3 5 10 **B.** 6 10 20 **C.** 9 25 100 **D.** Error

Answer

Let's evaluate one call at a time!

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`square(3)` = **9** `square(5)` = **25**

`square(10)` = **100**

Answer: C

9 25 100

Remember: Each function call is evaluated separately.

Functions Can Return Different Types

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A function does not always return an `int`!

The return type depends on what the function gives back.

Example 1 — `float`

```
float average(float a, float b)
{
    return (a+b)/2;
}
```

`float`

Return type

Example 2 — `char`

```
char grade(int marks)
{
    if(marks >= 90)
        return 'A';

    return 'B';
}
```

`char`

Return type

The first word tells us what type of value comes back.

The General Form of a Function

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So far, we have seen the pattern...

```
return_type function_name(parameters)
```

```
return_type function_name(parameters)
{
    statements;
    return value;
}
```

`return_type` `name` `(parameters)`

Example:

```
int square(int x)
```

`int` = return type `square` = name `int x` = parameter

What Is return?

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```
int add(int a, int b)
{
    return a + b;
}
```

`return a + b;` does **TWO things**

1. STOP
Function execution ends

→

2. SEND
Value goes back

Example

```
int x = add(10, 20);
```

`add(10,20)` → **30** → `x = 30`

`return` = **STOP + SEND BACK**

Can a Function Return Nothing?

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Yes!

```
void hello()
{
    printf("Hello Students!");
}
```

`hello()`
prints a message

`Hello Students!`

What does `void` mean?

`void` = **This function does not return a value.**

```
hello();
```

Output: `Hello Students!`

void Function

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```
void greet()
{
    printf("Good Morning!");
}
```



Notice something important:

```
int x = greet();    // X
```

Why is this wrong?

void Function

```
void greet()
{
    printf("Good Morning!");
}
```



Notice something important:

```
int x = greet();    // X
```

Why is this wrong?

Because `greet()` does not return a value.

Remember the return types

`int`	`float`	`char`	`void`
integer	float	character	nothing

return in a void Function

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```
void fun()
{
    printf("Hello");
    return;
}
```



What does `return;` mean here?

Stop executing the function and go back to the caller.

But look carefully...

```
void fun()
{
    return 10;      // wrong
}
```

Why?

`void` **means: no value is returned.**

The Big Difference

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`int add()`
returns a value

→

30
value comes back

`void greet()`
returns nothing

—

Hello!
just performs an action

Think of it like this:

Function → **Does work** → **Result?**

If there is a result to send back → use a return type.
If there is no result → use `void`.

Now Let's Solve a Problem

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We want to swap two numbers.

Initially

```
a = 10
b = 20
```



After swapping

```
a = 20
b = 10
```

Let's create a function:

```
void swap(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

Looks correct!

But...

Will it work?

Calling swap()

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```
int main()
{
    int a = 10;
    int b = 20;
    swap(a,b);
    printf("a = %d\n", a);
    printf("b = %d\n", b);
    return 0;
}
```

Predict the output!

A.
a = 20, b = 10

B.
a = 10, b = 20

C.
Error

D.
Random values

Answer: 10 20

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The output is:

```
a = 10
b = 20
```

But why?

```
a = b;
b = tmp;
```

Shouldn't the values change?

The secret:

The function received **copies** of the values.

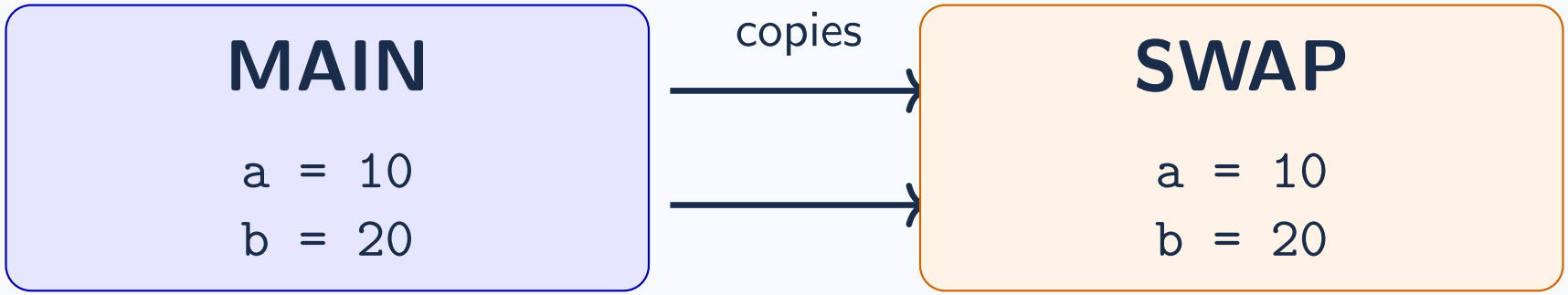
Call-by-Value

What Does Call-by-Value Mean?

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```
int a = 10;
int b = 20;
swap(a,b);
```

When `swap(a,b)` is called:



Important:

The function gets its **own variables** containing copies of the values.

So when `swap()` changes its `a` and `b`...
it changes only its copies.

Simple Example of Call-by-Value

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```
void change(int x)
{
    x = 100;
}

int main()
{
    int x = 10;
    change(x);
    printf("%d", x);
    return 0;
}
```

What is the output?

Simple Example of Call-by-Value

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```
void change(int x)
{
    x = 100;
}

int main()
{
    int x = 10;
    change(x);
    printf("%d", x);
    return 0;
}
```

What is the output?

10

Why?

main's x
x = 10



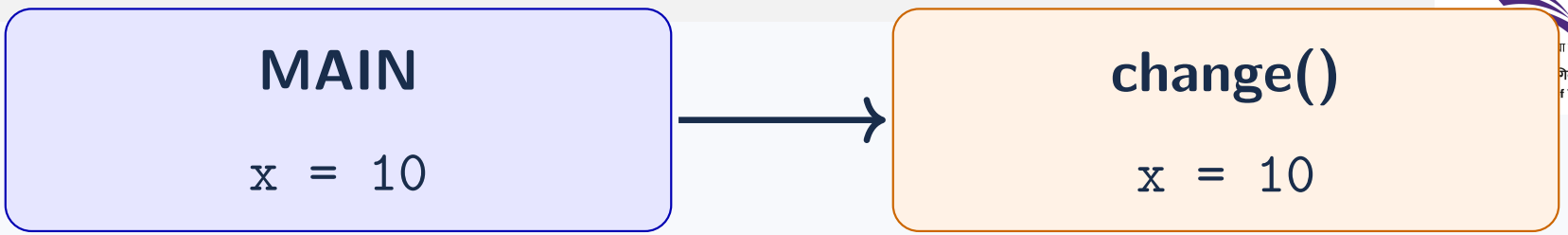
function's x
x = 10

The function changes its own copy:

x = 10 ⟶ x = 100

main's x is still 10!

Interactive Memory Question



Before calling:

```
main:
x = 10
```

Call: `change(x);`

During the function:



If the function executes:

```
x = 100;
```

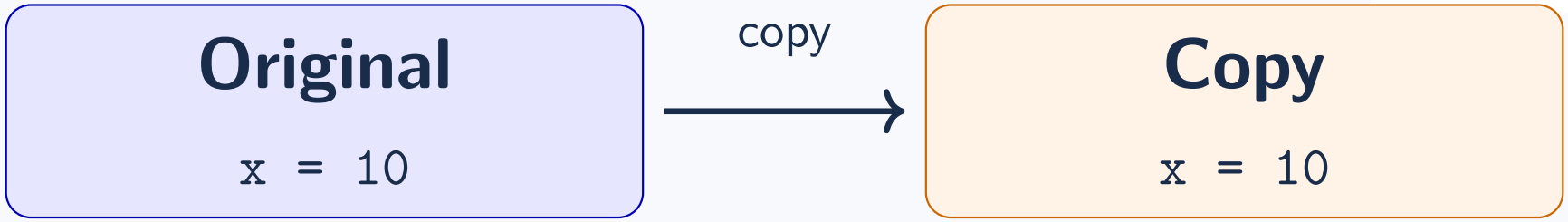
Which `x` changes?

Only the function's `x`.

After the function finishes: `main: x = 10`

The Key Idea: Copies, Not Originals

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Function changes: `copy: 10 → 100`

Original remains: `original: 10`

CALL-BY-VALUE

When we pass a variable to a function, the function receives a **copy of its value**.

This is why our first `swap()` did not work.

Why Didn't Our `swap()` Work?

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The function changes its own copies.

The original variables are untouched!

So how can the function change the ORIGINAL variables?

Why Didn't Our `swap()` Work?

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The function changes its own copies.

The original variables are untouched!

So how can the function change the ORIGINAL variables?

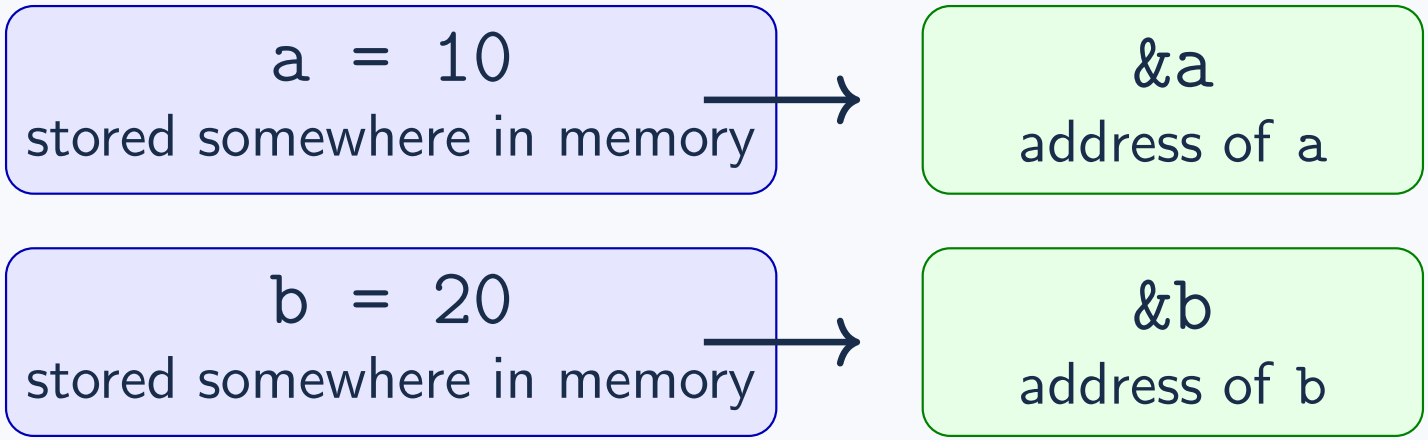
Give the function their addresses!

The Idea: Give the Function the Address

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```
int a = 10;
int b = 20;
swap2(&a, &b);
```

What do &a and &b mean?



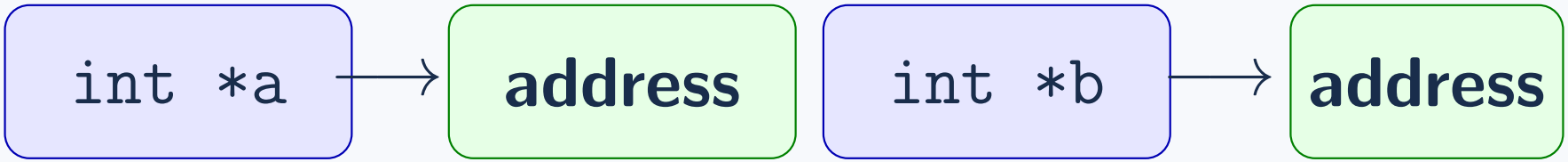
& means: **“Give me the address.”**

A New Version of `swap()`

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```
void swap2(int *a, int *b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
    return;
}
```

Look carefully at the parameters:



The function does **not** receive the numbers directly.

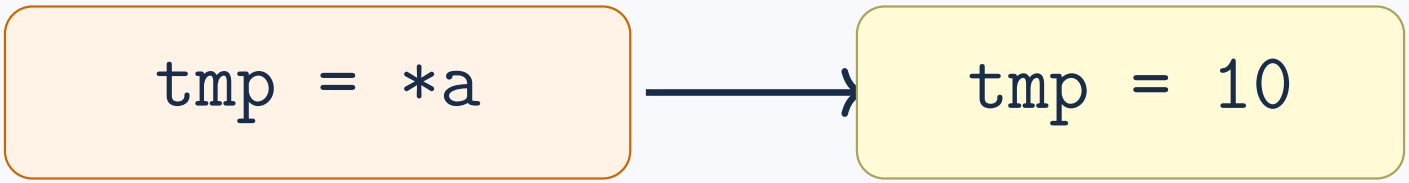
It receives their addresses.

How Does swap2() Actually Work?

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Now:



*a = *b ⇒ main's a = 20

*b = tmp ⇒ main's b = 10

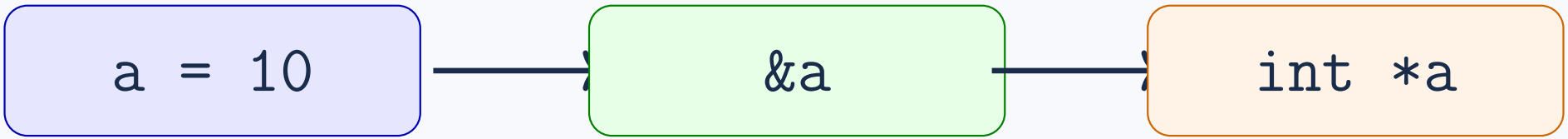
Calling swap2()

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```
int main()
{
    int a = 10;
    int b = 20;
    swap2(&a, &b);
    printf("a=%d b=%d", a, b);
    return 0;
}
```

What happens at:

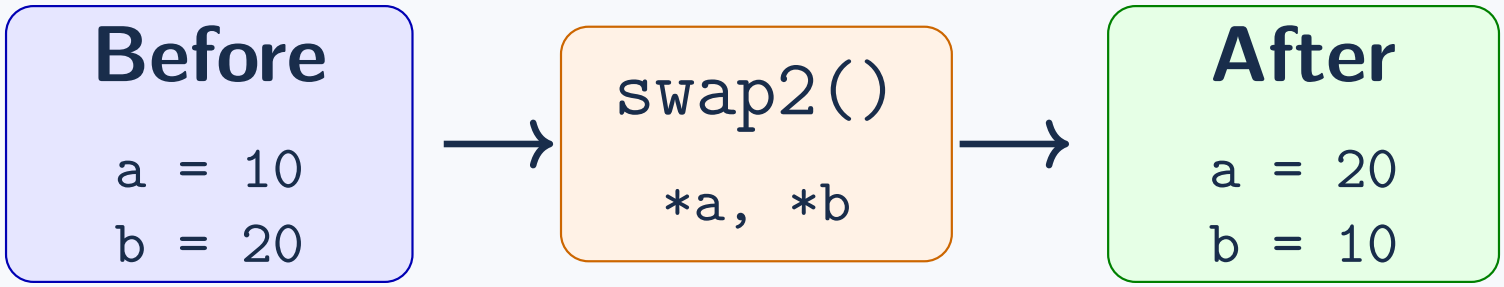
```
swap2(&a, &b);
```



The address of the original variable is passed.

Therefore, the function can reach the original variable.

Now the Swap Actually Works!



Why did it work?

We passed the **addresses** of the original variables.

The function used * to access those original variables.

This is commonly called Call-by-Reference.

Call-by-Value vs Call-by-Reference

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Call-by-Value

Pass the **value**

↓

Function gets a **copy**

↓

Original variable
does not change

Call-by-Reference

Pass the **address**

↓

Function can access
the **original variable**

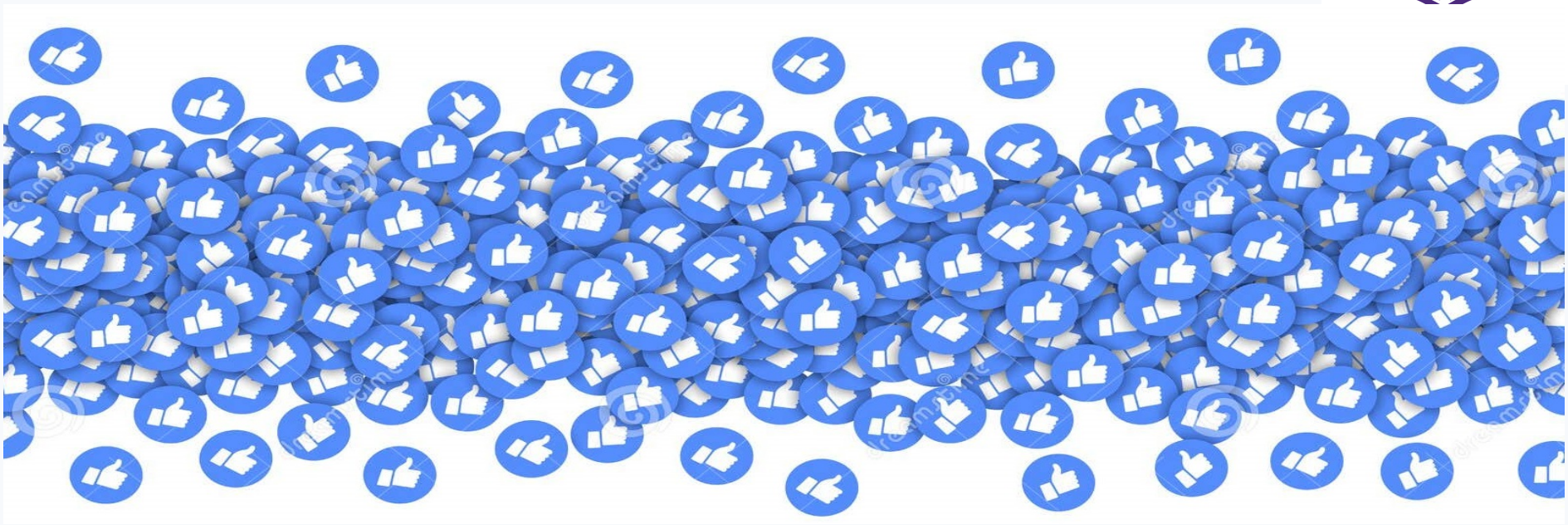
↓

Original variable
can change

Value: `swap(a,b)`

Reference: `swap2(&a,&b)`

Address + pointer = access to the original variable



Thank you for your kind attention!!